# Ajay Venkatesh

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## Skills

**Programming Languages:** C++, Java, Python, TypeScript, JavaScript (ES6), SQL
**Web Technologies:** HTML, CSS, React, REST API, GraphQL API, Microservices, Node.js, Express.js, Sequelize
**Database:** RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse
**Cloud:** AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs)
**Tools:** Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Kubernetes, Power BI, QuickSight, Backend system architecture, Database optimization, Cloud environments AWS, Cloud environments Google Cloud
**AI/ML:** LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, RL.

## Experience

### Campus Mesh — Jan 2026 – Present

**Software Engineer** | *Real-time microservices, recommendation engine, and LLM-based AI retrieval pipelines*

- Developed and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.
- Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing user engagement by **28%** and retention by **50%**.
- Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.

### Amazon — May 2025 – Aug 2025

**Software Development Engineer Intern** | *Serverless AWS infrastructure, React 18 frontend, and Java backend systems*

- Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the **system design** for approval and established Role Based Access **(4+ roles)** using **OAuth, Midway SSO, and HELIS**.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD deployments**.
- Built a **React 18** frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by **40%**.
- Engineered backend services in **Java** using **Coral, Smithy, Dagger**, and **CBOR serialization**; applied low-level design and design patterns, reduced backend execution time by **35%** and ensured reliability by unit and integration testing using **Mockito**.

### New York University (Novel AI Technologies, Inc.) — Aug 2024 – Present

**Software Developer (On-Campus)** | *NSF-funded AI health platform, ETL pipelines, and full-stack insurance analytics*

- Architected an **NSF-funded ($100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models for health metrics analysis and food image processing.
- Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
- Developed an insurance analytics platform **(React, Node.js, REST APIs)**; deployed **Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue.
- Streamlined underwriting via broker portal integrated with insurer platform; implemented **document and media processing** and **WebSocket-based real-time communication**; secured **6 insurance clients**.

### PricewaterhouseCoopers (PwC) — Aug 2021 – Aug 2024

**Senior Software Engineer** | *Cloud-native backend, NLP/OCR automation, and data engineering on Azure*

- Developed a distributed backend on **Microsoft Azure** for a health platform, integrating with **Microsoft CRM** and implementing event-driven processing and fault tolerance.
- Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.
- Deployed an **NLP based OCR neural model** on Azure to automate structured data extraction, reducing data processing time by **75%**.
- Optimized data ingestion with **parallel processing** for **100K**-row from Excel; reduced execution time from **5 hours to 1 hour**.
- Built **Power BI dashboards** processing large datasets from multiple databases, improving reporting efficiency by **40%**.

## Projects

**Advanced Expense Tracker (Java Desktop Application)** | ***Java, JavaFX, JDBC, SQLite, JFreeChart***

- Architected a modular **Java application** supporting individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns**.
- Implemented concurrent transaction handling and real-time cost splitting using **multithreading** and synchronized blocks to ensure thread-safe database updates.

**Hospital Management System (Full-Stack Application)** | ***React, Node.js, Express, MongoDB, SageMaker, Hugging Face***

- Built a **full-stack healthcare platform** using **React, Node.js, Express, and MongoDB** to manage appointments, patient records, lab reports, and medical history with **role-based access control**.
- Fine-tuned a **Hugging Face transformer-based model** achieving **88% accuracy** for health condition prediction and deployed it on **AWS SageMaker** for real-time diagnostic support in the doctor dashboard.

## Education

**New York University** | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** — Sep 2024 – May 2026

- *Coursework:* System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

## Certifications

**AWS Cloud Practitioner**
Azure Data Fundamentals
Microsoft Solution Architect Expert
Power BI Data Analyst Associate
Microsoft Developer Associate
Stanford Machine Learning (Coursera)